

Fundamentals of Natural Language Processing

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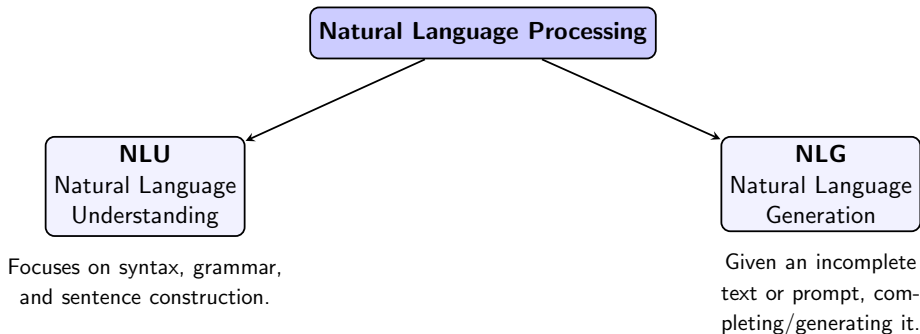
Outline

- 1 Introduction to Natural Language Processing (NLP)
- 2 Framing NLP as a Machine Learning Problem
- 3 Corpus and Tokenization
- 4 Text Pre-processing Techniques
- 5 Text Normalization: Stemming vs. Lemmatization
- 6 Vocabulary Building & Handling OOV Tokens
- 7 Brief Introduction to Featurization

What is Natural Language Processing?

Natural Language: Human languages like English, Hindi, French, etc.

NLP: Processing natural language so that we can learn useful features and automate language-based tasks.



Common Use Cases & Core NLU Tasks

NLP powers many modern applications. Here are standard implementations mapping raw text to actionable insights:

End-to-End Use Cases

Sentiment Analysis

Predicts (+ve / -ve / neutral) sentiment of a review.

Document Classification

Routing emails into 'Spam' vs 'Inbox'.

Document Clustering

Grouping similar news articles automatically.

Core NLU Extraction Tasks

Parts of Speech (POS) Tagging

Identifying nouns, verbs, adjectives.

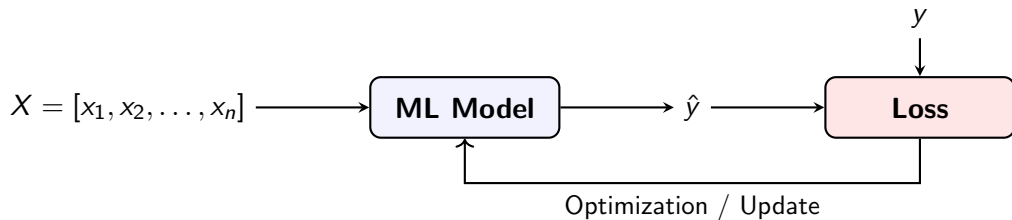
Named Entity Recognition (NER)

Extracting proper nouns (Names, Locations, Organizations).

Revisiting Traditional Supervised ML

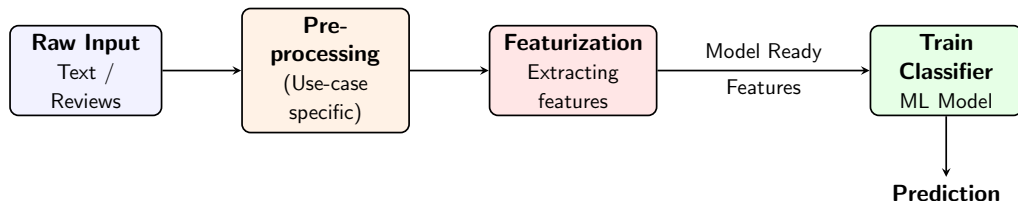
Before dealing with text, let's recall standard Machine Learning:

- **Features (X):** Usually numerical or categorical (encoded via One-Hot Encoding).
- **Targets (y):** The ground truth labels.



The Standard NLP Pipeline

How do we apply ML to raw text? We must transform text into numerical features through a standardized pipeline.



The fundamental first step of any NLP problem is transforming raw text to model-ready features.

Defining a Corpus

What is a Corpus?

A **Corpus** is a designated set of texts intended to be used for a specific NLP task.

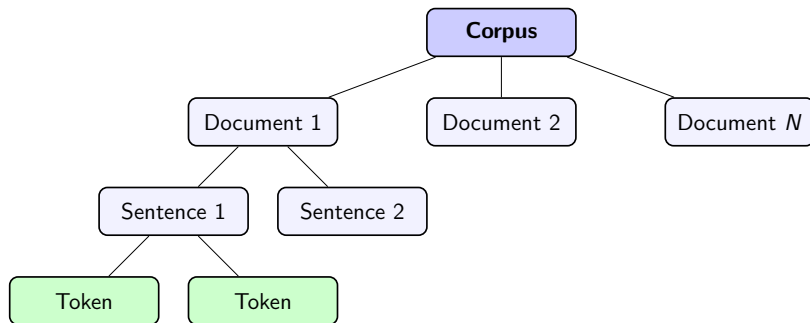
Examples:

- A set of Wikipedia articles.
- A dataset of product reviews from Amazon.
- Financial Times articles published between 2020 – 2025.

Data Collection Methods: One of the most common techniques to collect a corpus is **web-scraping** (crawling different websites and dumping textual data). Note that most sophisticated NLP models are trained on publicly available corpora.

Hierarchy of Text and Tokenization

Tokenization is the process of breaking down our corpus into its fundamental building blocks.



Tokens are the building blocks of a sentence. Note: In a strict sense, tokens can consist of a *part* of a word (e.g., 'telegraph' → 'tele' + 'grap').

Text Pre-processing: Removing Noise & Stopwords

Step 1: Removing Noise

- Stripping out punctuation: semicolons (;), colons (:), commas (,), emojis (:-)).

Step 2: Removing Stopwords

Stopwords are highly common words that carry little semantic weight for the specific task.

Language Specific

'the', 'and', 'is', 'are', 'on'

Domain Specific

If building a sentiment classifier for an air-fryer, the word 'air-fryer' appears everywhere and becomes uninformative noise.

$$\text{Total Stopwords} = \text{Language Specific} \cup \text{Domain Specific}$$

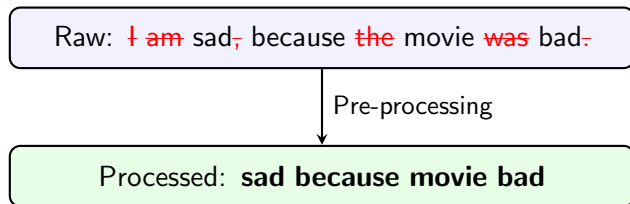
Text Pre-processing: Web Artifacts & Lowercasing

Step 3: Removing Web Artifacts

- Remove URLs (`http://...`) and handles (`@username`).

Step 4: Lowercasing

- Converting all text to uniform lowercase.
- Why? To the computer, 'Apple' and 'apple' are completely different features. Lowercasing reduces the dimensionality/sparsity of our feature set.



Text Normalization

Text Normalization is the process of reducing a word to its base or root form. There are two primary techniques: **Stemming** and **Lemmatization**.

1. Stemming

- A **crude heuristic process** that chops off word endings (prefixes/suffixes) to find the root.
- It does *not* care about grammar or meaning.
- Often produces non-dictionary words.
- Example algorithm: The Porter Stemmer for English.

Raw Word	Stemmed Form
Caring	car
Cars	car
Studies	studi

2. Lemmatization

- Uses vocabulary and morphological analysis of words to return the **lemma**.
- The lemma is the true, base dictionary word.
- It heavily takes into account the **Parts of Speech (POS)** and context.

Raw Word	Lemmatized Form
Caring	care
Studies	study
Better	good

Trade-offs: Stemming vs. Lemmatization

Stemming

- Fast and computationally cheap.
- Chops prefixes/suffixes crudely.
- Outputs non-dictionary words.
- No context awareness.

Use Case: Large search engines where speed > accuracy.

Lemmatization

- Computationally heavy / slow.
- Uses deep morphological rules.
- Guaranteed valid dictionary word.
- Highly context aware (POS).

Use Case: Chatbots and NLU tasks needing high accuracy.

Creating the Vocabulary (V)

In NLP, the **Vocabulary** (V) is the complete set of unique tokens present in the corpus after pre-processing.

Original Text	Pre-processed Text
I love to read horror stories	love read horror story
I hate the Engineering stream	hate engineer stream
I love my country	love country

Vocabulary (V): Set of Unique Words

$$V = \{ \text{love, read, horror, story, hate, engineer, stream, country} \}$$

$$|V| = 8$$

The size $|V|$ determines the number of features (dimensions) in our NLP model — every word in V becomes one dimension of the feature vector.

Handling Out-of-Vocabulary (OOV) Tokens

The Problem:

What happens if our model sees a word in production that wasn't in the training Vocabulary (V)?

Training Vocabulary: $V = \{\text{apple, orange, banana}\}$

Unseen Prediction Input: "I ate a **mango**"

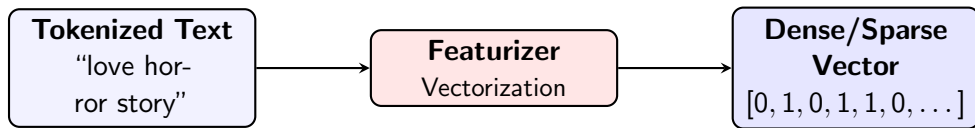
The Traditional Solution:

During training, we map extremely rare words to a universal unknown token, often denoted as $\langle \text{UNK} \rangle$.

- We update our vocabulary: $V = \{\text{apple, orange, banana, } \langle \text{UNK} \rangle\}$
- When the model encounters 'mango' during prediction, it maps it to $\langle \text{UNK} \rangle$ rather than crashing.
- The model learns a generic weight for $\langle \text{UNK} \rangle$ tokens based on how rare words behave in the training set.

From Text to Numbers: Featurization

Once we have our clean Tokens and our defined Vocabulary (V), we reach the final step before the Machine Learning algorithm: **Featurization**.



Techniques (to be covered in detail later):

- **Bag of Words (BoW):** Counting the frequency of vocabulary words in a document.
- **TF-IDF:** Weighing words based on their rarity across the whole corpus.
- **Embeddings:** Dense vectors mapping semantic meaning.

Thank You

Questions?